Appendix E. The Mathematics of Composing SLOs

When composing SLOs, you can use a statistical model to calculate the best possible availability of your service based on the availability of its dependencies. This model is determined by the architectural pattern used to call the dependencies.

Serial Dependencies

In a serial dependency chain, a request succeeds only if *all* calls in the sequence succeed.

Assume you have a Service A that calls Service B and then Service C. Their availability SLOs can be stated as probabilities of failure:

- Service B has 80% availability, so its failure probability is:

$$p_{fail(B)} = 1 - 0.80 = 0.2$$

- Service C has 90% availability, so its failure probability is:

$$p_{fail(C)} = 1 - 0.90 = 0.1$$

Assuming their failures are statistically independent, the probability of a successful call to Service A is the product of the success probabilities of its dependencies:

$$P(A_{success}) = P(B_{success}) \times P(C_{success})$$

$$P(A_{success}) = (1 - p_{fail(B)}) \times (1 - p_{fail(C)})$$

Plugging in our numbers:

$$P(A_{success}) = (1 - 0.2) \times (1 - 0.1) = 0.8 \times 0.9 = 0.72$$

This means the highest possible availability for Service A is 72%.

Parallel Dependencies (with Hedging)

In a parallel hedging pattern, a request fails only if *all* parallel calls fail.

Use the same Services B and C. The probability that Service A fails, $p_{fail(A)}$, is the probability that both B and C fail. Assuming independent failures, you can calculate this by multiplying their individual failure probabilities:

$$p_{fail(A)} = p_{fail(B)} \times p_{fail(C)}$$

Plugging in our numbers:

$$p_{fail(A)} = 0.2 \times 0.1 = 0.02$$

This gives Service A a failure probability of only 2%. The resulting availability for Service A is:

$$P(A_{success}) = 1 - p_{fail(A)} = 1 - 0.02 = 0.98$$

With this architecture, Service A can achieve 98% availability.

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